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Hybrid AI Handout — Apply + Evaluate on Your Project

Intellectual question: Does making source eligibility explicit improve your project's evidence—or simply encode a brittle rule?

Today: Test one bounded project function using baseline similarity retrieval versus metadata-aware eligibility. Submit a before/after evidence table, failure diagnosis, and ADOPT / MODIFY / REJECT / DEFER decision.

Before you run the test

Design element	Team specification
Project function	When a new charge comes in, decide which stored label the app is allowed to use as evidence: one that belongs to the student, is still current, and was confirmed by them.
Sources + IDs	Real records: L-012 (ALDI → Groceries), L-056 (Amazon Prime → Subscriptions), L-007 (Venmo → Transfers), L-039 (Planet Fitness → Health), L-199 (Urgent Care → Health). Tempting but not allowed: RM-001 (roommate's Venmo → Rent), V-056a (old Amazon Prime label → Shopping), BK-039 (bank's guess, Planet Fitness → Fees). Simulated: RM-001, V-056a, BK-039, the blank field on L-199, and the five October test charges.
Explicit semantics	source_id; status (current / superseded); authority (student_confirmed / bank_auto / blank); scope (student / roommate); effective_date; owner (student / roommate / nobody). The full list is saved in label_memory_snapshot.csv.
Rule + ownership	Use a record only if status = current, authority = student_confirmed, and scope = student. A blank field counts as no. The app sets status when a label is edited, scope from the account, and authority only when the student confirms. The student owns their own labels; our team owns the import code.
Success + failure	Should qualify: the student's current, confirmed labels. Unsafe: using a roommate's, an old, or an unchecked label. Not suitable if it keeps dropping good records because a field is missing.

Test design: Keep the project question, source texts, similarity method, and evaluation criteria fixed within each A/B comparison. A = unfiltered relevance ranking; B = the SAME ranking with your explicit eligibility rule. If you change a source's metadata for a stress test, run BOTH A and B on that same altered snapshot. Log excluded IDs and reasons. A manual/standard-library prototype is acceptable; label any simulated results.

Project Hybrid AI evidence table

For each case, predict first. Preserve source IDs, metadata snapshot, baseline top result, filtered top result (or no eligible result), and the reason for exclusion. The cases below are tests, not A/B conditions.

Test / question	Snapshot / expected eligibility	A: baseline top ID + why	B: eligible top ID / excluded IDs	Outcome / risk
1 · Normal eligible case ALDI $58.14 (Groceries)	3 ALDI records, all allowed	L-012 (1.00, first of 3 ties) → Groceries. Right.	L-012 / nothing excluded → Groceries. Right.	Same result. The rule costs nothing here.
2 · Archived / stale Amazon Prime $7.49 (Subscriptions)	V-056a superseded; L-056, L-128, L-205 allowed	V-056a (1.00, tied, saved first), the old label → Shopping. Wrong.	L-056 / excluded V-056a (status = superseded) → Subscriptions. Right.	Avoided an old label (S2).
3 · Wrong scope / audience Venmo $27.50 (Transfers)	RM-001 is the roommate's; the student's Venmo records are allowed	RM-001 (1.00, saved first) → Rent & Utilities. Wrong.	L-007 / excluded RM-001 (scope = roommate) → Transfers. Right.	Avoided someone else's label (S3).
4 · Missing / uncertain authority Planet Fitness $15.00 (Health)	BK-039 has a blank authority field; L-039 and later records allowed	BK-039, the bank's guess, saved before the student reviewed it → Fees. Wrong.	L-039 / excluded BK-039 (authority blank) → Health. Right.	Avoided an unchecked label. "CLUB FEES" fooled the bank the same way it fooled our A2 keyword rule.
5 · False authority OR valid source over-filtered Urgent Care $35.00 (Health)	L-199 is a correct label, but its scope field is blank	L-199 (1.00) → Health. Right.	Nothing left / excluded L-199 (scope blank). The next matches are paychecks at 0.19, so it asks the student.	A good record was lost (S4). Safe, since it only asks the student, but avoidable.

Evidence rule: For case 3, source RM-001 ranked 1st in A, but B excluded it because scope = roommate under our rule. This was justified because it's the roommate's rent payment, and for our student a Venmo payment is a transfer. Case 5 went the other way: L-199 was excluded for a blank scope field even though its label was right.

Decision + Project Dossier Update

Diagnose the semantic-layer failure (use codes below)

Code	Failure layer	What to look for
S1	False authority	A source is incorrectly marked official/authorized; unsafe inclusion.
S2	Stale status	Old source is still marked current, producing an obsolete result.
S3	Wrong scope	Evidence from a different course, audience, region, role, or task is used.
S4	Over-filtering	A legitimate source is excluded due to missing, rigid, or wrong metadata.
S5	Ungrounded output	An answer claims more than the eligible retrieved evidence supports.
S6	Maintenance	No accountable owner checks effective dates, revokes status, or repairs fields.

Count what happened (across five cases)

Measure	Team result
Eligibility decisions justified by source evidence	4 / 5. We checked each one against the charge text, note and our own labels, not just the metadata. Case 5 was not justified.
Unsafe inclusion avoided	3 / 3 relevant stress cases. Denominator: the cases where a disallowed record ranked first in A (2, 3 and 4).
False exclusions / no-eligible-source outcomes	1 case (case 5). Was it an appropriate abstention? Safe, but not really appropriate: the evidence was there and only a field was blank.
Observed A/B effect	Filtering improved the old-label, wrong-account and unchecked-label cases (A got 2/5 right, B got 4/5). It made case 5 worse and didn't change case 1.
Rule maintenance	Field verifier: the app's import and label-editing code (our team). Update owner: the student for their labels, our team for import bugs. Review trigger: any record with a blank field, a new import source, or a student edit.

Architecture comparison + project decision

Would explicit semantics be worth the complexity for this bounded function?

Alternative	When it may be preferable	Team note
Similarity / search only	When a human reviews evidence and source eligibility is not consequential.	Picked the wrong record in 3 of 5 cases, because tied records are taken in the order they were saved.
Metadata + eligibility rule	When verifiable source authority, currency, or scope constrains use.	Worked in our tests, but our app won't store the records it filters out, so the extra fields aren't worth it .
Human review / no automated answer	When authority is disputed, metadata absent, or exclusion errors are costly.	Best for records with blank fields, and for mixed stores where only the student knows what they bought.

Required team decision: REJECT

Decision sentence: We will reject a semantic eligibility layer for choosing which stored labels count as evidence for a new charge, because although in cases 2, 3 and 4 the baseline selected an old label, a roommate's label and the bank's guess while filtering picked the student's own label, our app won't store those kinds of records in the first place. The main unresolved risk is labels wrongly marked as confirmed, like an auto-accepted LLM guess (1 of 59 "sure" answers was wrong in September); our next validation is to check that saving only labels the student confirms, and replacing a label when it's edited, avoids the same three failures without a filter.

Project Dossier entry

Technique / bounded function: Similarity search plus a metadata rule, used to pick which stored label the app can trust for a new charge.

Source representation + metadata schema: 214 label records (211 of our real labels and 3 test records), each with cleaned text, amount, category, note, status, authority, scope, effective_date and owner.

Rule, verifier, owner + justification: A record is allowed only if it's current, confirmed by the student, and from their own account; a blank field fails. The app's code checks the fields and the student owns their labels. Only the student knows what they actually bought.

Baseline vs filtered test + case-level evidence: Same ranking in both. A got 2 of 5 right, B got 4 of 5. B excluded V-056a (old label), RM-001 (roommate), BK-039 (bank's guess) and L-199 (blank field).

Failure label + false inclusion/exclusion: No bad records got through in the five cases. One good record was dropped (S4, case 5). An extra test showed the rule would trust a wrong LLM label that was auto-accepted (S1).

Comparison to RAG-only / search-only + decision rationale: The filter only helps when the saved labels are a mix of good and bad records. Our app is for one student, doesn't import bank categories, and saves a label only when the student confirms it, so the records it filters out shouldn't exist. Keeping the saved labels clean is simpler than filtering them, and the rule also threw out a good record over a blank field.

Next test before deployment: Rerun the five cases on a snapshot with only confirmed labels and edits that overwrite old ones, and make sure auto-accepted LLM guesses are never saved as confirmed.